[2026] Machine Learning Projects

The objective of the projects is to prepare you to apply different machine learning algorithms to real-world tasks. This will help you to increase your knowledge about the workflow of the machine learning tasks. You will learn how to clean your data, applying pre-processing, feature engineering, regression, and classification methods. Each project will be delivered in milestones.

- The best three teams for each project will be honored.
- Registration starts: Monday 13/4/2026 7:00 PM.
- Registration ends: Wednesday 14/4/2026 at 11:59 PM.
- Delivering Milestone 1: 25/4/2026 11:59 PM.
- Delivering Milestone 2: Practical exam.
- Minimum number of members is 4 and the maximum is 6
- You must deliver a detailed report for each milestone containing all your work (feature analysis, algorithms used in each module and the achieved accuracy for each one)

Note : Each report will be graded

In the first milestone, you will apply the followings :-

Preprocessing: Before building your models, you need to make sure that the dataset is clean and ready-to-use.

Regression: Apply different regression techniques (at least two) to find the model that fits your data with minimum error.

Milestone 1: 50%

- Preprocessing, Regression.

Milestone 1 Report Must Include:

- You must explain in detail the **preprocessing techniques** you needed to apply on your dataset and how you implemented them.
- Perform **analysis** on the dataset as studied and explain how the features affect and relate to each other.
- You must explain what **regression techniques** you used (at least two).
- Mention the **differences** between each model and the acquired **results** (accuracy/error and so on).
- You must clearly mention **what features** you used or discarded to create your regression models.
- Explain what the **sizes** of your training, testing and validation sets are, if exist.
- Mention any further techniques that were used to **improve** the results (if exist).
- You should include **screenshots** of the resultant(s) regression line plots if applicable.
- Finally, write a **conclusion** about this phase of the project and what intuition you had about your problem and how it was proved/disproved.

Project(1)-SC-1&-AI-Rob-1: Online Games Popularity Prediction

Can you predict how popular an online game will be? This project explores a dataset containing information about various online games, such as genre, platform and more. By analyzing these factors, the goal is to identify the key drivers of game popularity and build a machine learning model that can estimate future **popularity** for new or existing games.

Dataset Snapshot:

QueryID	ResponseI	QueryNam	ResponseI	ReleaseDa	RequiredA	DemoCou	Developer	DLCCount	Metacritic	MovieCou	PackageC	Recomme	PublisherC	Screensho	SteamSpy	SteamSpy	SteamSpy	SteamSpy
10	10	Counter-S	Counter-S	Nov 1 200	0	0	1	0	88	0	1	68991	1	13	13033334	92789	9140731	78136
20	20	Team Fort	Team Fort	Apr 1 1999	0	0	1	0	0	0	1	2439	1	5	5399140	60368	753627	22699
30	30	Day of Def	Day of Def	May 1 200	0	0	1	0	79	0	1	2319	1	5	7621102	71499	1709740	34145
40	40	Deathmat	Deathmat	Jun 1 2001	0	0	1	0	0	0	1	888	1	4	7498965	70936	789676	23235
50	50	Half-Life: (	Half-Life: (	Nov 1 199	0	0	1	0	0	0	1	2934	1	5	5399499	60370	705741	21968
60	60	Ricochet	Ricochet	Nov 1 200	0	0	1	0	0	0	1	1965	1	4	7488563	70888	843481	24011
70	70	Half-Life	Half-Life	Nov 8 199	0	0	1	1	96	0	1	12486	1	11	5927504	63206	2268772	39303
80	80	Counter-S	Counter-S	Mar 1 200	0	0	1	0	65	0	1	7067	1	8	10489261	83543	4127910	52878
80	80	Counter-S	Counter-S	Mar 1 200	0	0	1	0	65	0	1	7067	1	8	10489261	83543	4127910	52878
130	130	Half-Life: E	Half-Life: E	Jun 1 2001	0	0	1	0	71	0	1	2219	1	5	5368650	60199	662517	21286
220	220	Half-Life 2	Half-Life 2	Nov 16 20	0	1	1	1	96	2	2	35792	1	9	9901173	81234	5695963	61979
240	240	Counter-S	Counter-S	Nov 1 200	0	0	1	0	88	0	1	53931	1	5	15433570	100626	11472993	87251

~Dataset header Continued:

Achievem	Achievem	Controller	IsFree	FreeVerAv	PurchaseA	Subscripti	PlatformW	PlatformLi	PlatformM	PCReqsHa	PCReqsHa	LinuxReqs	LinuxReqs	MacReqsH	MacReqsH	CategoryS	CategoryM	CategoryC
0	0	FALSE	FALSE	FALSE	TRUE	FALSE	TRUE	TRUE	TRUE	TRUE	FALSE	TRUE	FALSE	TRUE	FALSE	FALSE	TRUE	FALSE
0	0	FALSE	FALSE	FALSE	TRUE	FALSE	TRUE	TRUE	TRUE	TRUE	FALSE	TRUE	FALSE	TRUE	FALSE	FALSE	TRUE	FALSE
0	0	FALSE	FALSE	FALSE	TRUE	FALSE	TRUE	TRUE	TRUE	TRUE	FALSE	TRUE	FALSE	TRUE	FALSE	FALSE	TRUE	FALSE
0	0	FALSE	FALSE	FALSE	TRUE	FALSE	TRUE	TRUE	TRUE	TRUE	FALSE	TRUE	FALSE	TRUE	FALSE	FALSE	TRUE	FALSE
0	0	FALSE	FALSE	FALSE	TRUE	FALSE	TRUE	TRUE	TRUE	TRUE	FALSE	TRUE	FALSE	TRUE	FALSE	TRUE	TRUE	FALSE
0	0	FALSE	FALSE	FALSE	TRUE	FALSE	TRUE	TRUE	TRUE	TRUE	FALSE	TRUE	FALSE	TRUE	FALSE	FALSE	TRUE	FALSE
0	0	FALSE	FALSE	FALSE	TRUE	FALSE	TRUE	TRUE	TRUE	TRUE	FALSE	TRUE	FALSE	TRUE	FALSE	TRUE	TRUE	FALSE
0	0	FALSE	FALSE	FALSE	TRUE	FALSE	TRUE	TRUE	TRUE	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	TRUE	TRUE	FALSE
0	0	FALSE	FALSE	FALSE	TRUE	FALSE	TRUE	TRUE	TRUE	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	TRUE	TRUE	FALSE
0	0	FALSE	FALSE	FALSE	TRUE	FALSE	TRUE	TRUE	TRUE	TRUE	FALSE	TRUE	FALSE	TRUE	FALSE	TRUE	FALSE	FALSE
33	10	FALSE	FALSE	FALSE	TRUE	FALSE	TRUE	TRUE	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	TRUE	FALSE	FALSE
147	10	FALSE	FALSE	FALSE	TRUE	FALSE	TRUE	TRUE	TRUE	TRUE	FALSE	TRUE	FALSE	TRUE	FALSE	FALSE	TRUE	FALSE

~Dataset header Continued:

CategoryM	CategoryIr	CategoryIr	CategoryIr	CategoryV	GenreIsNo	GenreIsInc	GenreIsAc	GenreIsAd	GenreIsCa	GenreIsStr	GenreIsRP	GenreIsSir	GenreIsEa	GenreIsFre	GenreIsSp	GenreIsRa	GenreIsMa	PriceCurre
FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	USD
FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	USD
FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	USD
FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	USD
FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	USD
FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	USD
FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	USD
FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	USD
FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	USD
FALSE	FALSE	TRUE	FALSE	FALSE	FALSE	FALSE	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	USD
FALSE	FALSE	TRUE	FALSE	FALSE	FALSE	FALSE	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	USD

~Dataset header Continued:

PriceCurre	PriceInitia	PriceFinal	SupportEn	SupportUF	AboutText	Backgroun	ShortDesc	DetailedD	DRMNotic	ExtUserAc	HeaderIma	LegalNotic	Reviews	Supported	Website	PCMinReq	PCRecReq	LinuxMinR
USD	9.99	9.99		http://stea	Play the w	http://cdn.		Play the w			http://cdn.			English Fre	None	Minimum:		Minimum:
USD	4.99	4.99			One of the	http://cdn.		One of the			http://cdn.			English Fre	None	Minimum:		Minimum:
USD	4.99	4.99			Enlist in ar	http://cdn.		Enlist in ar			http://cdn.			English Fre	http://www	Minimum:		Minimum:
USD	4.99	4.99			Enjoy fast-	http://cdn.		Enjoy fast-			http://cdn.			English Fre	None	Minimum:		Minimum:
USD	4.99	4.99			Return to t	http://cdn.		Return to t			http://cdn.			English Fre	None	Minimum:		Minimum:
USD	4.99	4.99			A futuristic	http://cdn.		A futuristic			http://cdn.			English Fre	None	Minimum:		Minimum:
USD	9.99	9.99		http://stea	Named Ga	http://cdn.		Named Ga			http://cdn.			English Fre	http://www	Minimum:		Minimum:
USD	9.99	9.99		http://stea	With its ex	http://cdn.		With its ex			http://cdn.			English Fre	None	Minimum:		
USD	9.99	9.99		http://stea	With its ex	http://cdn.		With its ex			http://cdn.			English Fre	None	Minimum:		
USD	4.99	4.99			Made by G	http://cdn.		Made by G			http://cdn.			English Fre	None	Minimum:		Minimum:
USD	9.99	9.99		http://stea	1998. HAL	http://cdn.	#app_220_	1998. HAL			http://cdn.			English* Fi	http://www			
USD	19.99	19.99		http://stea	THE NEXT I	http://cdn.	Just updat	THE NEXT I			http://cdn.			English Fre	None	Minimum:		Minimum:

Milestone 1 tasks:

1. Apply pre-processing on the provided dataset. (You must preprocess all the features even if you won't use them later after feature selection).
2. Apply Feature Selection and Experiment with regression techniques to reduce the error on prediction of the "RecommendationCount" (Deliver at least two regression models with significant difference).
3. Finish Milestone 1 Report.

Note: You must preprocess all features, but model and feature selection can be done after that (i.e You can drop a feature only after preprocessing and with valid reason).

Project(2)-SC-2: E-Learning Student Performance Prediction

Can you predict how well students will perform across multiple online courses based on their personal characteristics and learning behavior? This project explores a dataset containing student information such as gender, region, age band, number of previous course attempts, prior education, and more. By analyzing these features, the goal is to identify the key factors influencing academic performance and build a machine learning model capable of predicting **student scores** across different subjects.

This dataset has seven .csv files that you need to combine and use:
assessments.csv, courses.csv, studentAssessments.csv, studentinfo.csv, studentRegistration.csv, studentVle.csv,vle.csv

Dataset Snapshot (assessments.csv):

code_mod	code_pres	id_assessi	assessme	date	weight
AAA	2013J	1752	TMA	19	10
AAA	2013J	1753	TMA	54	20
AAA	2013J	1754	TMA	117	20
AAA	2013J	1755	TMA	166	20
AAA	2013J	1756	TMA	215	30
AAA	2013J	1757	Exam	?	100
AAA	2014J	1758	TMA	19	10
AAA	2014J	1759	TMA	54	20
AAA	2014J	1760	TMA	117	20
AAA	2014J	1761	TMA	166	20
AAA	2014J	1762	TMA	215	30
AAA	2014J	1763	Exam	?	100

Dataset Snapshot (courses.csv):

code_mod	code_pres	module_presentation_length
AAA	2013J	268
AAA	2014J	269
BBB	2013J	268
BBB	2014J	262
BBB	2013B	240
BBB	2014B	234
CCC	2014J	269
CCC	2014B	241
DDD	2013J	261
DDD	2014J	262
DDD	2013B	240
DDD	2014B	241

Dataset Snapshot (studentAssessments.csv):

id_assessi	id_student	date_subm	is_banked	score
1752	11391	18	0	78
1752	28400	22	0	70
1752	31604	17	0	72
1752	32885	26	0	69
1752	38053	19	0	79
1752	45462	20	0	70
1752	45642	18	0	72
1752	52130	19	0	72
1752	53025	9	0	71
1752	57506	18	0	68
1752	58873	19	0	73
1752	59185	18	0	67

Dataset Snapshot (studentinfo.csv):

code_mod	code_pres	id_student	gender	region	highest_education	imd_band	age_band	num_of_pr	studied_cr	disability
AAA	2013J	11391	M	East Anglian Region	HE Qualification	90-100%	55<=	0	240	N
AAA	2013J	28400	F	Scotland	HE Qualification	20-30%	35-55	0	60	N
AAA	2013J	30268	F	North Western Region	A Level or Equivalent	30-40%	35-55	0	60	Y
AAA	2013J	31604	F	South East Region	A Level or Equivalent	50-60%	35-55	0	60	N
AAA	2013J	32885	F	West Midlands Region	Lower Than A Level	50-60%	0-35	0	60	N
AAA	2013J	38053	M	Wales	A Level or Equivalent	80-90%	35-55	0	60	N
AAA	2013J	45462	M	Scotland	HE Qualification	30-40%	0-35	0	60	N
AAA	2013J	45642	F	North Western Region	A Level or Equivalent	90-100%	0-35	0	120	N
AAA	2013J	52130	F	East Anglian Region	A Level or Equivalent	70-80%	0-35	0	90	N
AAA	2013J	53025	M	North Region	Post Graduate Qualific	?	55<=	0	60	N
AAA	2013J	57506	M	South Region	Lower Than A Level	70-80%	35-55	0	60	N
AAA	2013J	58873	F	East Anglian Region	A Level or Equivalent	20-30%	0-35	0	60	N

Dataset Snapshot (studentRegistration.csv):

code_module	code_presentat	id_student	date_registration	date_unregistration
AAA	2013J	11391	-159	?
AAA	2013J	28400	-53	?
AAA	2013J	30268	-92	12
AAA	2013J	31604	-52	?
AAA	2013J	32885	-176	?
AAA	2013J	38053	-110	?
AAA	2013J	45462	-67	?
AAA	2013J	45642	-29	?
AAA	2013J	52130	-33	?
AAA	2013J	53025	-179	?
AAA	2013J	57506	-103	?
AAA	2013J	58873	-47	?

Dataset Snapshot (studentVle.csv):

code_mod	code_pres	id_student	id_site	date	sum_click
AAA	2013J	28400	546652	-10	4
AAA	2013J	28400	546652	-10	1
AAA	2013J	28400	546652	-10	1
AAA	2013J	28400	546614	-10	11
AAA	2013J	28400	546714	-10	1
AAA	2013J	28400	546652	-10	8
AAA	2013J	28400	546876	-10	2
AAA	2013J	28400	546688	-10	15
AAA	2013J	28400	546662	-10	17
AAA	2013J	28400	546890	-10	1
AAA	2013J	28400	547011	-10	1
AAA	2013J	28400	547013	-10	1

Dataset Snapshot (vle.csv):

id_site	code_mod	code_pres	activity_type	week_from	week_to
546943	AAA	2013J	resource	?	?
546712	AAA	2013J	oucontent	?	?
546998	AAA	2013J	resource	?	?
546888	AAA	2013J	url	?	?
547035	AAA	2013J	resource	?	?
546614	AAA	2013J	homepage	?	?
546897	AAA	2013J	url	?	?
546678	AAA	2013J	oucontent	?	?
546933	AAA	2013J	resource	?	?
546708	AAA	2013J	oucontent	?	?
546995	AAA	2013J	resource	?	?
546884	AAA	2013J	url	?	?

Milestone 1 tasks:

1. Apply pre-processing on the provided dataset. (You must preprocess all the features even if you won't use them later after feature selection)
2. Apply Feature Selection and Experiment with regression techniques to reduce the error on prediction of the "score" column (Deliver at least two regression models with significant difference).
3. Finish Milestone 1 Report.

Note: You must preprocess all features, but model and feature selection can be done after that (i.e You can drop a feature only after preprocessing and with valid reason)

Project(3)-CS-1: Software Developers Job Satisfaction Prediction

Can you predict how satisfied software developers are with their jobs based on their tools, industry, and work practices? This project explores a dataset containing information such as platforms used, industry sector, adoption of AI tools, along with other developer responses to a survey. By analyzing these features, the goal is to identify the key drivers of job satisfaction and build a machine learning model that can predict satisfaction levels among software developers.

Dataset Snapshots:

Response	MainBran	Age	EdLevel	Employm	Employm	WorkExp	LearnCod	LearnCod	LearnCod	AILearnHc	YearsCode	DevType	OrgSize	ICorPM	RemoteW	Purchasel	TechEndo	TechEndo	TechEndo
1	I am a dev	25-34 year	Masterâ€	Employed	Caring for	8	Yes, I am r	Online Co	Yes, I learn	AI CodeGe	14	Develope	20 to 99 er	People ma	Remote	Yes, I influ	Work	10	7
2	I am a dev	25-34 year	Associate	Employed		2	Yes, I am r	Online Co	Yes, I learn	AI CodeGe	10	Develope	500 to 999	Individual	Hybrid (sc	No	Personal F	13	1
3	I am a dev	35-44 year	Bachelorâ	Independ	None of th	10	Yes, I am r	Online Co	Yes, I learn	AI CodeGe	12	Developer, front-end				No	Work	12	2
4	I am a dev	35-44 year	Bachelorâ	Employed	None of th	4	Yes, I am r	Other onli	Yes, I learn	AI CodeGe	5	Develope	10,000 or r	Individual	Remote	No	Personal F	2	12
6	I am a dev	45-54 year	Masterâ€	Independ	Caring for	15	Yes, I am r	Online Co	Yes, I learn	Technical	20	Developer, back-end				Yes, I influ	Work	12	5
7	I am a dev	25-34 year	Some coll	Independ	None of th	9	Yes, I am r	Videos (ne	Yes, I learn	AI CodeGe	13	Developer, full-stack				No	Personal F	11	1
8	I am a dev	35-44 year	Professior	Employed	Engaged i	22	Yes, I am r	Books / P	Yes, I learn	AI CodeGe	30	Architect,	Less than	People ma	Remote	Yes, I influ	Personal F	12	2
9	I am a dev	25-34 year	Bachelorâ	Employed	Caring for	9	Yes, I am r	Online Co	Yes, I learn	Technical	15	Data engir	5,000 to 9,	Individual	Remote	Yes, I influ	Personal F	5	7
10	I am a dev	25-34 year	Masterâ€	Employed	None of th	7	Yes, I am r	Other onli	No, I learned sometl		10	Develope	20 to 99 er	Individual	Remote	No	Work	13	2
11	I am a dev	25-34 year	Masterâ€	Employed	None of th	7	No, I am not new to		Yes, I learn	AI CodeGe	9	Develope	100 to 499	Individual	Remote	Yes, I influ	Work	11	4
12	I am a dev	18-24 year	Bachelorâ	Employed	None of th	3	No, I am not new to		Yes, I learn	Videos (ne	5	Develope	100 to 499	Individual	Remote	No			
13	I am a dev	35-44 year	Bachelorâ	Independ	Caring for	10	Yes, I am r	Videos (ne	No, I learned sometl		10	Developer, front-end				No	Personal F	6	13
14	I am a dev	25-34 year	Bachelorâ	Employed	Caring for	6	Yes, I am r	Online Co	Yes, I learn	AI CodeGe	6	Develope	1,000 to 4,	People ma	In-person	Yes, I influenced the		9	2
15	I am a dev	35-44 year	Professior	Employed	Attending	15	No, I am not new to		No, I learned something that w			Architect,	20 to 99 er	People ma	Remote	No	Personal F	9	7
16	I am a dev	18-24 year	Bachelorâ	Student	Engaged i	5	Yes, I am r	Online Co	No, I learned sometl		3	Develope	20 to 99 er	Individual	Remote	Yes, I end	Work	9	3
17	I am not p	65 years o	Some coll	Retired	None of th	20	No, I am not new to		No, I didn't spend tir		40	Financial analyst or engineer					Personal F	7	12
18	I am a dev	25-34 year	Masterâ€	Employed		1	Yes, I am r	Online Co	Yes, I learn	Videos (ne	8	Develope	100 to 499	Individual	Hybrid (sc	Yes, I end	Work	12	3
19	I am a dev	35-44 year	Associate	Employed		20	Yes, I am r	Stack Ove	No, I didn't spend tir		20	Develope	20 to 99 er	People ma	Hybrid (sc	Yes, I influ	Work	14	6
20	I am not p	25-34 year	Masterâ€	Independ	Engaged i	3	Yes, I am r	Videos (ne	Yes, I learn	AI CodeGe	6	Academic	I donâ€™t	Individual	Hybrid (sc	No	Personal F	12	2

~Dataset header Continued:

TechEndo	TechEndo	TechEndo	TechEndo	TechEndo	TechEndo	TechEndo	TechEndo	TechEndo	TechOppc	TechOppc	TechOppc	TechOppc	TechOppc	TechOppc	TechOppc	TechOppc	TechOppc	TechOppc	TechOppc
9	6	3	11	12	1	8	14		15	7	8	12	11	1	6	13	3	16	
2	9	4	3	12	5	7	14		14	1	13	10	9	3	4	2	5	15	
3	7	5	10	13	1	4	14		8	1	2	12	13	3	9	4	11	15	
6	5	13	3	8	11	7	14		10	5	8	4	13	12	2	7	1	15	
4	6	2	10	3	7	8	14		12	5	13	10	9	1	4	6	2	15	
2	6	4	7	13	8	3	14		8	3	14	11	4	1	10	5	13	15	
1	10	4	5	8	6	7	14		14	3	4	9	1	5	2	11	12	15	
13	4	9	8	12	2	11	14		11	13	10	1	5	9	7	8	2	15	
5	1	6	9	10	7	4	14		14	1	9	10	8	5	2	3	12	15	
5	13	7	8	2	3	10	14												
11	1	2	10	7	9	5	14												
8	13	11	6	3	10	5	14		3	8	5	13	14	10	2	7	6	15	
4	11	3	8	2	12	6	14		2	6	10	3	1	5	9	12	8	15	
6	7	2	8	1	4	5	14		7	4	8	14	5	12	3	13	9	15	
2	11	5	10	6	9	3	14	usefullne	4	10	5	1	12	13	3	2	11	15	
4	5	1	2	9	7	11	14		11	6	9	14	8	2	5	1	4	15	
5	1	2	4	12	3	10	11												
3	13	5	7	8	4	6	14		9	13	4	3	1	14	10	6	2	15	

~Dataset header Continued:

Industry	AIThreat	NewRole	ToolCoun	ToolCoun	Country	Currency	CompTota	Language	Language	Language	Language	Languages	Languages	Database	Database	Database	Database	Database	Database
Fintech	I'm not su	I have nei	7	3	Ukraine	EUR Europ	52800	Yes	Bash/Shel	Dart	Dart			Yes	Cloud Firestore;PostgreSQL				
Retail and	I'm not su	I have trar	6	5	Netherlar	EUR Europ	90000	Yes	Java	Java;Pyth	Java			Yes	Dynamod	Dynamod	Dynamodb;MongoDB		
Software	No	I have trar	3	3	Ukraine	UAH Ukrai	2214000	Yes	Dart;HTM	Dart;HTM	Dart;HTML/CSS;JavaScript;Type			Yes	MongoDB	Firebase F	MongoDB		
Retail and	No	I have neither consider or trans			Ukraine	EUR Europ	31200	Yes	Java;Kotli	Java;Kotli	Java;Kotlin			No					
Fintech	I'm not su	I have son	5	5	Ukraine	USD Unite	120000	Yes	Java;Scala	Java;Kotli	Java;Scala			No					
Insurance	Yes	I have somewhat considered cl			Ukraine	USD Unite	6240	Yes	JavaScript	JavaScript	JavaScript;TypeScript			No					
Software	Yes	I have stro	10	10	Ukraine	USD Unite	72000	Yes	Bash/Shel	Assembly	Bash/Shell (all shells);HTML/CS			Yes	MongoDB	PostgreSC	PostgreSQL;SQLite		
Banking/F	No	I have transitioned into a new			Ukraine	USD Unite	70000		Java;Pyth	Scala	Scala				Cassandra	Dynamod	Dynamodb;MySQL;PostgreSQL		
Software	No	I have son	6	2	Ukraine	USD Unite	5400	Yes						No					
Fintech	No	I have nei	9	4	Ukraine	UAH Ukrai	2740980	Yes	HTML/CSS	HTML/CSS	HTML/CSS;Java;JavaScript;SQL;			Yes	H2;Micros	H2;Micros	H2;Microsoft SQL Server;Postgr		
Fintech	No	I have neither consider or trans			Ukraine	USD Unite	14400	Yes	C#	C#	C#			Yes	Microsoft	Microsoft	Microsoft SQL Server		
Software	I'm not su	I have nei	2	1	Ukraine	UAH Ukrai	45840		HTML/CSS	C#;TypeSc	TypeScript			No					
Fintech	No	I have strongly considered char			India	INR Indian rupee								Yes					
Fintech	I'm not su	I have stro	5	2	Georgia	USD United States d		No						Yes	MySQL	MySQL	MySQL		
Software	Yes	I have somewhat considered cl			India	INR Indian rupee		Yes	Ada;Asser	Ada;Bash/	Ada;C;C#;	5	1	Yes	BigQuery;	Dynamod	InfluxDB	5	4
				1	Australia	AUDAustr	25000	Yes				MQL4	MQL5	No					
Governme	Yes	I have son	10	20	Greece	EUR European Euro		Yes	HTML/CSS	Python	Python			Yes	MariaDB;M	MongoDB	MongoDB		
Fintech	No	I have son	2	1	Germany	EUR European Euro		Yes	Bash/Shel	Bash/Shel	Bash/Shell (all shells);Java			Yes	H2;Mongo	MongoDB	MongoDB;PostgreSQL		
Energy	No	I have stro	2	2	Banglades	BDTBangladeshi taka		Yes	Python	Python	Python			No					

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PlatformC	PlatformH	PlatformV	PlatformA	PlatformH	PlatformV	Webfram	Webfram	Webfram	Webfram	Webfram	Webfram	DevEnvsC	DevEnvsH	DevEnvsW	DevEnvsA	DevEnvHa	DevEnvW	SOTagsHa	SOTagsWa
Yes	Amazon Web Services (AWS);Cloudflare;Firebase;G					No						Yes	Android Studio;Notepad++;Visual Studio;Visual Studio Code;Xcode						
Yes	Amazon V	Amazon V	Amazon Web Services (AWS);D			Yes	Spring Bo	Spring Bo	Spring Boot			Yes	IntelliJ IDI	IntelliJ IDI	IntelliJ IDEA;PyCharm;Visual Studio Code;Xcode				
Yes	Datadog;F	Amazon V	Firebase;npm;pnpm			Yes	Next.js;N	Next.js;N	Next.js;Node.js;Rea		Solid.js	Yes	Visual Stu	Visual Stu	Visual Studio Code			Google Ge	Google Ge
Yes	Amazon V	Amazon V	Amazon Web Services (AWS);G			Yes	Spring Bo	Spring Bo	Spring Boot			No						Amazon Bedrock	
Yes	Amazon V	Amazon V	Amazon Web Services (AWS)			Yes	Spring Bo	Spring Bo	Spring Bo	akka	akka	Yes	IntelliJ IDI	IntelliJ IDI	IntelliJ IDEA;Visual Studio Code				
Yes	Amazon Web Services (AWS);Docker;Homebrew;np					Yes	Next.js;N	Next.js;N	Next.js;Node.js;React			Yes	WebStorn	WebStorn	WebStorm				
Yes	Amazon V	APT;Cargo	APT;Docker;Homebrew;Pip			Yes	Angular;D	Django	Django			Yes	PyCharm;	Claude Co	PyCharm;Visual Studio Code			Amazon B	Odoo;Larg
	Amazon V	Amazon V	Amazon Web Services (AWS);Datadog;Mi			FastAPI;Spring Boot							IntelliJ IDEA;Notepad++;PyCharm;Visual Studio Code						
Yes						No						Yes	Xcode	Cursor					
Yes	Kubernete	Kubernete	Kubernetes;Maven (build tool)			Yes	Spring Bo	Spring Bo	Spring Boot			Yes	Cursor;Int	Cursor;Int	Cursor;IntelliJ IDEA;Notepad++			RAG	RAG
No						Yes	ASP.NET C	ASP.NET C	ASP.NET Core			No							
No							Angular	Angular;N	Angular			No							
No						No						No						Google Gemini	
Yes	Amazon V	Ansible;C	Pip	5	4	Yes	Angular;A	Drupal;Fa	Drupal;Re	4	4	No						c++23;Amazon Bedro	
No						No						No							
No						Yes	Django;Fl	Django;Re	Django;React			Yes	Vim;Visua	Vim;Visua	Vim;Visual Studio Code			Tailwind CSS 4	
Yes	Amazon V	Amazon V	Amazon Web Services (AWS);D			No						Yes	IntelliJ IDI	IntelliJ IDI	IntelliJ IDEA;Vim				
Yes	Google Cl	Google Cl	Google Cloud			No						Yes	Visual Stu	Visual Stu	Visual Studio Code			Google Ge	Large Lang

~Dataset header Continued:

SOTagsAd	SOTagsHa	SOTagsWa	OpSysPer	OpSysPro	OfficeStac	OfficeStac	OfficeStac	OfficeStac	OfficeStac	CommPla	CommPla	CommPla	CommPla	CommPla	AIModels	AIModels	AIModels	AIModels	AIModels
			Windows;	Windows;	Confluenc	Markdow	Markdown File			Stack Overflow;Reddit;GitHub (public projects, not					Yes	openAI GPT (chatbot models);openAI Ima			
			Windows;	MacOS	Confluenc	Confluenc	Confluence;GitHub;Jira			Stack Ove	Stack Ove	Stack Overflow;Discord;Reddit			Yes	openAI GI	openAI GI	openAI GPT (chatbot	
Google Gemini			MacOS;Ar	MacOS	GitHub;Gi	GitHub;Gi	GitHub;GitLab;Jira			Stack Ove	Stack Ove	Stack Overflow;Discord;Dev.to			Yes	Gemini (F	Gemini (F	Gemini (Flash genera	
			Windows	Windows;	GitLab;Jira	GitLab;Jira	GitLab;Jira			Stack Ove	Stack Ove	Stack Overflow;Reddit;YouTub			No				
			MacOS	MacOS	Azure Dev	GitHub;Jir	GitHub;Jira			Stack Ove	Stack Ove	Stack Overflow;Stack Exchange			No				
			MacOS;De	MacOS											No				
e Language Model;RAG			MacOS;iO	MacOS;Ub	GitHub;Jir	GitHub;Gi	GitHub;Jira			Stack Ove	Stack Ove	Stack Overflow;Stack Exchange			Yes	openAI GI	Anthropic	openAI GPT (chatbot	
			MacOS;iO	MacOS;iO	GitHub;Jira					Stack Ove	Stack Ove	Stack Overflow;GitHub (public			No				
RAG			Windows;	Windows	Confluenc	Confluenc	Confluence;GitHub;Obsidian;S			Stack Ove	Stack Ove	Stack Overflow;Reddit;GitHub			Yes	Anthropic	Anthropic	Anthropic: Claude Sc	
			Windows;	Windows	Confluenc	Confluenc	Confluence;GitHub;Jira			Discord;Y	Discord;Y	Discord;YouTube;Twitch			No				
			Windows	Windows	Azure Dev	Azure Dev	Azure Devops			Stack Ove	Stack Ove	Stack Overflow;YouTube;Medium				openAI GI	openAI GI	openAI GPT (chatbot	
			Windows;	Windows;	GitHub;Gi	GitHub;Gi	GitHub;GitLab			Discord	Discord	Discord			No				
ck;Microso	5	4	Windows;	Windows;	Airtable;A	Notion;Ob	Notion;Ob	4	5	Stack Overflow;Stack Exchange			5	5	Yes	Alibaba Cloud Qwen models;Amazon Tita			
	None of th	Not listed	Windows	Windows				No	No				None	None	No				
			MacOS	Ubuntu	GitHub;Google Colab;Jira					Stack Ove	GitHub (p	GitHub (public projects, not pri			Yes	openAI GPT (chatbot models)			
			Ubuntu;D	Ubuntu;D	GitHub;Gc	GitHub;Gc	GitHub;Google Workspace;Wik			Stack Ove	Stack Ove	Stack Overflow;Discord;Reddit			Yes	DeepSeek (R- Reasoning models);DeepSe			
uage Model			Windows	Windows	GitHub	GitHub	GitHub			Kaggle	Kaggle	Kaggle			No				

~Dataset header Continued:

AIModels'	SOAccoun	SOVisitFre	SODuratic	SOPartFre	SO_Dev_C	SO_Actior	SO_Actior	SO_Actior	SO_Actior	SO_Actior	SO_Actior	SO_Actior	SO_Actior	SO_Actior	SO_Actior	SOComm	SOFrictior	AISelect
ıge genera	Yes	A few tim	Between	I have never particip		1		6	2	3	7	8	9	11	15	Neutral	Rarely, alr	Yes, I use
: models)	Not sure/	Multiple t	Between	Infrequen	Chat (peo	6		5	8	2	4	9	10	11	15	Yes, some	About hal	Yes, I use
al purpose	Not sure/	A few tim	Between	Infrequen	Chat (peo	3		4	5	1	2	7	9	11	15	Neutral	About hal	Yes, I use
	Yes	A few tim	Between	I have nev	Videos;Ch	1		4	5	6	7	8	9	11	15	Neutral	Rarely, alr	Yes, I use
	Yes	A few tim	Between	Infrequen	Lists of re	6		9	3	4	5	10	1	11	15	Yes, some	Rarely, alr	Yes, I use
	Not sure/	A few tim	Between	Infrequen	Long-form articles;Videos;Lists of recommendations (tools, frameworks, technologies, etc.);Day-in-the-life profiles/interviews with experts;Cha													Yes, I use
: models)	Yes	A few tim	Between	Infrequen	Chat (peo	7		10	1	6	5	4	2	11	15	Yes, defin	About hal	Yes, I use
						1		4	5	6	7	8	9	11	15			Yes, I use
	Yes	Daily or al	Between	I have never participated in Q&A on Stack Overflow												No, not re	I don't use	No, and I
ɔnnet;oper	Yes	A few tim	Between	I have nev	Long-form	1		4	5	6	7	8	9	11	15	Yes, some	About hal	Yes, I use
	Yes	Less than	Between	Infrequently, less than once per year												No, not at	Less than	Yes, I use
: models)	Yes	A few tim	Between	Infrequen	Videos;Ch	2		5	1	6	7	8	9	11	15	No, not re	Less than	Yes, I use
	Yes	Less than	Less than one year															
	Yes	Multiple t	Between	A few tim	Chat (people);Coding challenges											Neutral	About hal	Yes, I use
n models;	Yes	Multiple t	Between	Less than	Message Boards;Videos;Coding challenges;Logic games/puzzles;Lists of recommendations (tools, frameworks, technologies,											Yes, defin	About hal	Yes, I use
	No	Less than	I don't use	I have nev	Other (wr	11		5	2	7	8	4	3	6	15	I don't use	No, not at	I don't use
	Yes	A few tim	Between	Infrequen	Chat (peo	1		5	8	2	11	10	3	14	15	No, not re	About hal	Yes, I use
ɜek (V- Ger	Yes	A few tim	Between	Less than	Long-form articles											No, not at	I don't use	No, and I
	Yes	Infrequen	Between	I have nev	Day-in-the	4		13	5	8	9	6	1	14	15	No, not at	I don't use	Yes, I use

~Dataset header Continued:

AISent	AIAcc	AIComple	AIToolCur	AIToolDor	AIToolPlai	AIToolPlai	AIToolCur	AIFrustrat	AIExplain	AIAgents	AIAgentCl	AIAgent_U	AgentUse	AIAgentIn	AIAgentIn	AIAgentIn	AIAgentIn	AIAgentIn	AIAgentCl
Indifferen	Neither tr	Bad at har	Learning a	Project pl	Documenting code;Debugging			AI solutio	No	Yes, I use	Not at all	Software engineerin		AI agents	AI agents have helped me automate repe				I am conce
Indifferen	Neither tr	Bad at har	Learning r	Learning a	Writing code;Documenting cod			AI solutio	No	No, and I	Not at all or minimally								It takes sig
Favorable	Somewha	Neither g	Documenting code;D	Project pl	Learning about a cod			AI solutio	No	Yes, I use	Yes, some	Software	Multi-plat	AI agents	AI agents	AI agents have improved collab			It takes sig
Favorable	Somewha	Bad at handling com		Project pl	Learning a	Search for answers;C		AI solutio	NO	Yes, I use	Not at all	Software	Language	AI agents	AI agents	AI agents have helped me autc			It takes sig
Indifferen	Somewha	Good, but	Learning a	Predictive	Deployme	Project pl	Testing co	AI solutio	No it isnt	Yes, I use	Not at all	Software engineerin		AI agents have incre		AI agents	AI agents have helpe		The cost o
Favorable	Somewha	Neither g	Learning about a cod		Project planning;Debugging or			AI solutions that are		Yes, I use	Yes, some	Software engineerin		AI agents	AI agents have improved the quality of m				The cost o
Very favo	Somewha	Neither g	Writing code;Docum		Project pl	Learning a	Search for	AI solutio	Yes	No, but I p	Yes, to a great extent								
Indifferen	Neither tr	Neither good or bad at handling complex tasks										Software engineering;Data and analytics							
Unfavorat	Highly dis	Very poor at handling complex		Testing code				AI solutio	No.	No, but I p	No, but my development work has changed somewhat due to non-AI factors								The cost o
Very favo	Somewha	Very poor	Learning a	Debugging or fixing code;Testi			Search for	AI solutio	When I	Yes, I use	Yes, some	Software engineerin		AI agents	AI agents have impr		AI agents have increased my pr		
Very favo	Neither tr	Bad at har	Learning a	Project planning;Documenting			Search for answers;Learning ne			No, but I p	Not at all or minimally								I have con
Unfavorat	Somewha	Bad at handling com		Project pl	Writing cc	Learning about a cod		lâ€™ve be	No	No, and I	No, but my development work has significantly changed due to non-AI factors								Write in th
Favorable	Somewha	Good, but	Learning about a codebase;Project planni				Document	AI solutions that are		No, and I	No, but my development work has significantly changed due to non-AI factors								I am conce
Favorable	Highly tru	Good, but	Search for answers			Document	Learning a	AI solutions that are		Yes, I use	Yes, to a g	Software	Language processing;Agent/multi-agent				AI agents have increased my pr		
Very unfa	Highly dis	I don't use AI tools f		Learning about a codebase;Project planni			I donâ€™t	No		No, and I	No, but my development work has significantly changed due to non-AI factors								
Unfavorat	Neither tr	Good, but	Writing cc	Project planning;Tes		Learning a	Generatin	AI solutions that are		No, and I	Yes, somewhat								Integratin
Very unfa	Highly dis	Very poor at handlin		Learning a	Writing code;Predictive analyti			Other (wr	No	No, and I	No, but my development work has changed somewhat due to non-AI factors								Integratin
Very favo	Neither tr	Very well at handling complex tasks					Learning a	AI solutio	yes	No, but I p	Yes, to a great extent								I am conce

~Dataset header Continued:

AIAgentCl	AIAgentCl	AIAgentCl	AIAgentCl	AIAgentKr	AIAgentKr	AIAgentO	AIAgentO	AIAgentO	AIAgentO	AIAgentE>	AIAgentE>	AIHuman	AIOpen	Convertec	JobSat	
Integratin	The cost of using certain AI agent platforms is a barr					Vertex AI				ChatGPT		When I dc	Troublesh	61256	10	
ɡnificant ti	I am conce	Integrating AI agents with my existing tools and workflows can be difficult.;My company's IT and/or Info										When I dc	All skills.	104413	9	
My company's IT and		I am concerned abou		Redis						ChatGPT;Claude Cod		When I dc	Understar	53061	8	
ɡnificant ti	I am conce	Integrating AI agents with my existing tools and workflows can be difficult.								ChatGPT;Claude Cod		When I donâ€™t trus		36197	6	
f using cer	I am conce	Integratin	My company's IT and/or InfoSec teams have strict rules that do not allow							GitHub Copilot		When I ha	designing	120000	7	
f using cer	I am conce	It takes significant time and effort to learn how to use AI agen						Sentry		ChatGPT;Claude Cod		When I donâ€™t trus		6240	6	
I have con	I am conce	The cost o	Integrating AI agents with my existing tools and workflows can be difficult.;My company's IT a									When I wa	Innovatio	72000	6	
														70000	8	
It takes sig	I have con	I am conce	My company's IT and/or InfoSec teams have strict rules that do not allow me to use AI agent t									When Iâ€™m stuck a		5400	4	
'oductivity	I am conce	Integratin	It takes significant time and effort to lear				Spring AI			ChatGPT;Claude Cod		When I dc	Code	65691	10	
cerns abou	I am conce	It takes sig	My company's IT and/or InfoSec teams have strict rules that do not allow me to use AI agent t									When I donâ€™t trus		14400	8	
ɜe most sig	Integratin	I am concerned about the accuracy of the information provided by AI agents.;It takes significant time an										When I want to fully		1099	7	
															6	
ɜrned about the accuracy of the information provided by AI agents.;Integrating AI agents with my existing tools and workflow												I donâ€™t senior development			9	
'oductivity	I am conce	The cost of using cer		Fireproof;LanceDB;L		Vertex AI;Agno;Autc		Galileo;Honeycomb;		ChatGPT;AgentGPT;A		When I dc	5		7	
	I am concerned about the accuracy of the information provided by AI agents.;Integrating AI agents with my existin											Other (ple	All the on	16252		
It takes sig	I have con	I am conce	My company's IT and/or InfoSec teams have strict rules that do not allow me to use AI agent t									When I want to learn best prac			6	
g AI agents	I am conce	My company's IT and/or InfoSec teams have strict rules that do not allow me to use AI agent tools or pla										When I dc	AI will not become n		5	
ɜrned abou	Integrating AI agents with my existing tools and workflows can be difficult.;It takes significant time and effort to le											I donâ€™t	Research			

Milestone 1 tasks:

1. Apply pre-processing on the provided dataset. (You must preprocess all the features even if you won't use them later after feature selection)
2. Apply Feature Selection and Experiment with regression techniques to reduce the error on prediction of the "JobSat Score" (Deliver at least two regression models with significant difference).
3. Finish Milestone 1 Report.

Note: You must preprocess all features, but model and feature selection can be done after that (i.e You can drop a feature only after preprocessing and with valid reason).

Project(4)-CS-2: Movie Popularity Prediction

Can you predict how popular a movie will be based on its characteristics and production details? This project explores a dataset containing information such as genres, production companies, runtime, spoken languages and more. By analyzing these features, the goal is to identify the key factors that influence movie **popularity** and build a machine learning model capable of estimating how well a film will perform with audiences.

Dataset header:

id	title	quality	theatrical	movie_sen	movie_val	movie_vad	movie_vad	movie_vad	movie_inte	movie_inte	movie_inte	movie_inte	movie_inte	movie_inte	movie_inte	movie_inte	movie_scl	movie_scl
565770	Blue Beetl	real_confic	TRUE	positive	0.434939	0.445939	0.070364	0.405667	0.058714	0.077	0.062571	0.2165	0.231857	0	0	0.209143	-0.061	0.266667
980489	Gran Turis	real_confic	TRUE	positive	0.508375	0.4949	0.2317	0.2794	0	0	0	0	0.347667	0	0	0.541833	0	0.333333
968051	The Nun II	real_confic	TRUE	negative	-0.13256	-0.15391	0.115182	0.198364	0.319818	0.097273	0.236455	0.365773	0	0.261364	0.117909	0.140636	0	0.461538
615656	Meg 2: The	real_confic	TRUE	neutral	-0.00315	0.100458	0.067333	0.172458	0	0.1465	0	0	0.0715	0.09775	0	0.1075	0	0.285714
762430	Retributior	real_confic	TRUE	negative	-0.3	-0.31833	0.316333	0.096667	0.713667	0	0.127667	0.749	0	0.680333	0.395667	0.044333	-0.00858	0.857143
1008042	Talk to Me	real_confic	TRUE	negative	-0.11056	-0.09292	0.209879	0.105379	0.264667	0.0268	0.183333	0.551867	0.069567	0.391633	0.1182	0.0719	-0.16725	0.434783
385687	Fast X	real_confic	TRUE	neutral	-0.0228	0.121143	0.234714	0.12	0.32475	0.297	0.15425	0.164	0.125	0.1835	0.1425	0	0	0.714286
678512	Sound of F	real_confic	TRUE	negative	-0.20029	-0.14138	0.320167	-0.00554	0.2109	0.1569	0	0.3475	0.1726	0.339	0.2298	0.1219	-0.1675	0.285714
346698	Barbie	real_confic	TRUE	positive	0.175524	0.187857	0.029976	0.233643	0.109	0.016917	0.083	0.048875	0.153125	0.072917	0	0.292	-0.02375	0.571429
976573	Elemental	real_confic	TRUE	positive	0.329583	0.447	-0.062	0.085	0	0.3515	0	0.318	0.2795	0	0.117	0.3165	0	0.714286
1002338	Operation	real_confic	TRUE	positive	0.27975	0.27975	-0.219	0.14375	0	0	0	0	0	0	0	0.43		0
820609	No One Wi	real_confic	TRUE	negative	-0.45131	-0.38363	0.131233	-0.06867	0.126313	0.05025	0.171	0.285125	0.060688	0.347188	0.036125	0.043938	-0.06458	0.6

~Dataset header Continued:

movie_scl	vote_av	vote_coun	status	release_date	revenue	runtime	adult	backdrop_	budget	homepage	imdb_id	original_la	original_tit	overview	popularity	poster_pat	tagline
0.266667	7.139	1023	Released	8/16/2023	124818235	128	FALSE	/1syW9SN	1.2E+08	https://ww	tt9362930	en	Blue Beetl	Recent col	2994.357	/mXLOHH	Jaime Rey
0.333333	8.068	702	Released	8/9/2023	114800000	135	FALSE	/xFYpUmB	60000000	https://ww	tt4495098	en	Gran Turis	The ultima	2680.593	/51tqzRtK	From gam
0.461538	6.545	365	Released	9/6/2023	231200000	110	FALSE	/53z2fXEKI	38500000	https://ww	tt1016097	en	The Nun II	In 1956 Fra	1692.778	/c9kVD7W	Confess y
0.285714	6.912	2034	Released	8/2/2023	384056482	116	FALSE	/5mzr6JZb	1.29E+08	https://ww	tt9224104	en	Meg 2: The	An explora	1567.273	/4m1Au3Y	Back for s
0.857143	6.816	258	Released	8/23/2023	12905464	91	FALSE	/iiXliCeykk	20000000	https://iiil.	tt6906292	en	Retributior	When a my	1547.22	/oUmmY7	All roads l
0.434783	7.214	973	Released	7/26/2023	72600000	95	FALSE	/ilvQnZyzg	4500000	https://um	tt1063852	en	Talk to Me	When a gr	1458.514	/kdPMUMJ	You call. T
0.714286	7.265	3881	Released	5/17/2023	704709660	142	FALSE	/4XM8DUT	3.4E+08	https://fas	tt5433140	en	Fast X	Over many	1175.267	/fiVW06jE	The end of
0.285714	7.973	503	Released	7/3/2023	212587173	131	FALSE	/eMPxmN	15000000	https://ww	tt7599146	en	Sound of F	The story c	1111.036	/kSf9svfL2	Fight for th
0.571429	7.279	5074	Released	7/19/2023	1428545028	114	FALSE	/ctMserH8	1.45E+08	https://ww	tt1517268	en	Barbie	Barbie and	1069.34	/iuFNMS8	She's even
0.714286	7.757	2467	Released	6/14/2023	486797988	102	FALSE	/4fLZUr1e	2E+08	https://mo	tt1578903	en	Elemental	In a city wh	1008.942	/4Y1WNkd	Opposites
0	7.04	25	Released	1/26/2023	0	112	FALSE	/oghHR3X	0		tt1548539	is	NapÃ³leon	A modern-	982.611	/j2Or0w69	A dark sec
0.6	6.897	345	Released	9/22/2023	0	93	FALSE	/zYlgqIpqJ	22800000	https://ww	tt1450911	en	No One Wi	An exiled a	961.212	/ehGIDAM	A home in
0.764706	7.321	56	Released	9/26/2023	29300000	118	FALSE	/9b3uxqcjl	13000000	https://ww	tt2180722	en	Saw X	Between th	949.967	/aQPeznS	Witness th

~Dataset header Continued:

genres	production_compani	production_coun	spoken_languages
Action, Science Fictic	Warner Bros. Picture	United States of	English, Portuguese,
Action, Drama, Adven	PlayStation Producti	United States of	English, German, Jap
Horror, Mystery, Thrill	New Line Cinema, At	United States of	English, French
Action, Science Fictic	Apelles Entertainmei	China, United Sta	English
Action, Mystery, Thrill	StudioCanal, The Pic	France, United S	English, German
Horror, Thriller	Causeway Films, Me	Australia, United	English
Action, Crime, Thriller	Universal Pictures, C	United States of	English
Action, Drama	Santa Fe Films	United States of	English, Spanish
Comedy, Adventure, F	LuckyChap Entertain	United Kingdom,	English
Animation, Comedy, F	Walt Disney Pictures	United States of	English
Thriller	Sagafilm, Splendid Fi	Germany, Icelan	English, Icelandic
Horror, Science Fictic	20th Century Studios	United States of	English

Milestone 1 tasks:

1. Apply pre-processing on the provided dataset. (You must preprocess all the features even if you won't use them later after feature selection)
2. Apply Feature Selection and Experiment with regression techniques to reduce the error on prediction of the "popularity" (Deliver at least two regression models with significant difference).
3. Finish Milestone 1 Report.

Note: You must preprocess all features, but model and feature selection can be done after that (i.e You can drop a feature only after preprocessing and with valid reason).

Project(5)-AI-Rob-2: Vitamin Deficiency Prediction

Can personal lifestyle choices and environmental factors accurately predict nutritional gaps and deficiencies? This project explores a comprehensive dataset containing demographic data, lifestyle habits, and dietary intake metrics, such as age, BMI, sun exposure, and daily RDA (Recommended Dietary Allowance) percentages for essential vitamins and minerals. By analyzing these multi-faceted features—ranging from geographic latitude to symptomatic history—the goal is to identify the primary drivers of nutritional health and develop a machine learning model capable of diagnosing potential vitamin deficiencies. This tool aims to provide an early-warning system for individuals to optimize their health through data-driven dietary adjustments.

Dataset Snapshot:

age	gender	bmi	smoking_	alcohol_c	exercise_	diet_type	sun_expo	income_le	latitude_r	vitamin_a	vitamin_c	vitamin_d	vitamin_e	vitamin_b	folate_pe	calcium_p	iron_perc	symptoms	symptoms
79	Male	24.8	Former	None	Active	Vegetaria	High	High	Mid	119.1	147.3	152.88	97.5	102.5	188.9	108.3	97.4	0	None
77	Female	39.9	Former	Moderate	Light	Omnivore	Low	Low	Low	85.7	57.5	32.76	82.7	62.6	51	42.6	102.5	1	bone_pai
24	Male	26.4	Former	Heavy	Moderate	Omnivore	Low	High	High	48.3	152.1	94.99	169.3	136.2	116.6	136.3	86.4	2	dry_skin;r
69	Male	23.1	Never	Heavy	Moderate	Vegetaria	High	Low	Low	75.8	51	51.48	85.7	31.8	66.5	76.5	60.8	2	numbness
63	Male	29.6	Never	None	Moderate	Vegetaria	Moderate	High	Low	93.3	111.5	62.9	155.6	72.6	124.9	69.4	71.9	0	None
69	Female	25.1	Never	Moderate	Active	Pescataria	Moderate	Low	Mid	47.9	75.3	65.5	72.2	21.2	101	70	79	3	numbness
31	Male	25.3	Former	Heavy	Light	Vegetaria	Low	Low	High	62.2	106.7	25.27	53.7	27	77.4	67.1	54.7	2	bone_pai
73	Female	28.1	Former	Heavy	Active	Vegetaria	Low	High	Mid	129.6	138	42.56	70.8	50	162.1	151.2	120.4	2	bone_pai
18	Male	30.4	Current	Moderate	Light	Pescataria	High	High	Low	192.5	123.5	166.01	84.7	125.9	98.1	198.9	40.3	1	fatigue
25	Male	18.4	Never	Moderate	Active	Pescataria	Moderate	Middle	Mid	111.2	108.3	111.8	67.8	88.7	104.7	89.7	77	0	None
40	Female	33.9	Never	Heavy	Light	Vegetaria	Moderate	Low	High	65.7	45.8	22.5	56.1	54.5	46.4	49.3	46.5	2	bone_pai
46	Male	28	Former	None	Active	Pescataria	High	High	Mid	109.5	230.5	223.21	81.1	110.6	120.6	134.2	84.7	0	None
28	Female	17.8	Former	Moderate	Sedentary	Omnivore	Moderate	High	Low	158.4	115.2	112.8	60	154.2	104.3	112.7	10	2	pale_skin;
49	Female	23.9	Current	None	Active	Omnivore	High	Low	Low	94.7	57.2	83.33	108.6	63.7	53.8	40.1	57	0	None
59	Male	15	Current	Heavy	Sedentary	Vegetaria	High	Low	High	62	55	53.82	18.9	56.4	48.3	53.3	45.4	1	pale_skin
51	Female	27.6	Never	None	Light	Pescataria	High	High	Low	116	64	127.53	135.2	129.5	45.5	106.3	135.7	0	None
64	Female	29.9	Current	None	Sedentary	Vegan	High	Low	High	41.5	45.3	45.37	81.8	25.3	46.5	62.9	54.4	2	bone_pai
19	Female	25.5	Never	None	Moderate	Vegan	High	Middle	Mid	76.5	157.8	84.24	104	27.4	142	36.9	40.9	3	numbness
19	Female	26.5	Former	Heavy	Light	Vegetaria	Low	Middle	High	118.4	57.7	38.43	42.7	76.5	123	73.8	56.5	2	bone_pai

~Dataset header Continued:

symptom:	symptoms	vitamin_deficiency
0	None	0
1	bone_pai	0.844
2	dry_skin;r	0.004
2	numbness	0.7724
0	None	0.372
3	numbness	0.82
2	bone_pai	1.27
2	bone_pai	0.8276
1	fatigue	0.04
0	None	0.08
2	bone_pai	1.1478
0	None	0
2	pale_skin;	0.19
0	None	0.079
1	pale_skin	0.6208
0	None	0
2	bone_pai	1.176
3	numbness	0.703
2	bone_pai	0.7326

Milestone 1 tasks:

1. Apply pre-processing on the provided dataset. (You must preprocess all the features even if you won't use them later after feature selection)
2. Apply Feature Selection and Experiment with regression techniques to reduce the error on prediction of the "vitamin_deficiency" (Deliver at least two regression models with significant difference).
3. Finish Milestone 1 Report.

Note: You must preprocess all features, but model and feature selection can be done after that (i.e You can drop a feature only after preprocessing and with valid reason).